

SIMATS ENGINEERING

Department of Computer Science & Engineering

CO3-AT3 Assignment

Programming in Java

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Register No.	192472012
Course Code	CSA0904
Course Name	Programming in Java
Assignment	CO3-AT3 Assignment



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Q1 Product Price Input NumberFormatException

25 marks

Problem Statement:
Develop a Java program that accepts the price of a product as a String and converts it into a numeric value. Handle invalid numeric input without terminating the program.

Requirements:

Read product price as a String.
Convert it using Integer.parseInt() or Double.parseDouble().
Handle NumberFormatException.
Display an appropriate error message.

Q3
Online Examination ♦ The
Synchronization
25 marks

Q4
Electricity Bill Calculation
Custom Exception
25 marks

4/4 answered

Your Code

```
1 import java.util.Scanner;
2
3 public class ProductPriceInput {
4     public static void main(String[] args) {
5         Scanner scanner = new Scanner(System.in);
6
7         String priceInput = scanner.nextLine();
8
9         try {
10
11             double price = Double.parseDouble(priceInput);
12             System.out.println("The product price is: " + price);
13         }
14     }
15 }
```

Input

500

Output

The product price is: 500.0

QUESTIONS

Q1

Product Price Input 

NumberFormatException

25 marks

Q2 

Online Ticket Booking 

Thread Synchronization

25 marks

Q3 

Online Examination  Thread Synchronization

25 marks

Q4 Electricity Bill Calculation
Custom Exception
25 marks

4/4 answered

Q2 Online Ticket Booking ♦ Thread Synchronization

25 marks

Problem Statement:

Develop a multithreaded Java program to simulate an online ticket booking system. Multiple users should attempt to book seats simultaneously.

Requirements:

Create multiple booking threads.

Maintain a common number of available seats.

Use synchronization to prevent multiple threads from booking the same seat.

Display successful and unsuccessful booking attempts.

Your Code

```
1 public class Main {
2     static int seats = 3;
3
4     static synchronized void bookSeat(String user) {
5         if (seats > 0) {
6             System.out.println(user + " booked a seat. Seats left: " +
7                 seats--);
8         } else {
9             System.out.println(user + " failed. No seats available.");
10        }
11    }
12 }
```

Input

```
stdin input
```

Output

```
User1 booked a seat. Seats left: 2
User4 booked a seat. Seats left: 1
User3 booked a seat. Seats left: 0
User2 failed. No seats available.
```


SIMATS Online Lab

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JAVA PROGRAMMING

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QUESTIONS

Q1
Product Price Input
NumberFormatException
25 marks

Q2
Online Ticket Booking
Thread Synchronization
25 marks

Q3
Online Examination
Thread Synchronization
25 marks

Q4
Electricity Bill Calculation
Custom Exception
25 marks

4/4 answered

Q4 Electricity Bill Calculation Custom Exception 25 marks

Problem Statement:
Develop a Java program to calculate an electricity bill based on units consumed. If the user enters a negative number of units generate a user-defined InvalidUnitsException.

Requirements:
Read consumer name and units consumed.
Validate the units.
Create and use a custom exception.
Calculate the bill only for valid input.
Handle the exception gracefully.

Your Code

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6
7         try {
8             System.out.print("Enter Consumer Name: ");
9             String name = sc.nextLine();
10
11             System.out.print("Enter Units Consumed: ");
12             int units = sc.nextInt();
13
14         } catch (InvalidUnitsException e) {
15             System.out.println(e.getMessage());
16         }
17     }
18 }
```

Input

Jeswanth
60

Output

Enter Consumer Name: Enter Units Consumed:
--- Electricity Bill ---
Consumer Name: Jeswanth
Units Consumed: 60
Total Bill Amount: 30.0

32°C
Mostly sunny

Search



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